Multimodal Sentiment Analysis Across Western and Bollywood Music: A Cross-Cultural Comparative Study Using Audio Features and Lyrics

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Abstract—Music is a universal language, but how we express "sadness" or "joy" depends on where we come from. Usually, people think sad words mean slow music. But in Bollywood, a heartbreak song might have a fast dance beat. This paper studies this "cultural mismatch." We used 18,194 Western songs and 222 Bollywood songs to see if AI can understand these differences. By looking at both the sound (using Spotify derived audio features) and the lyrics (using XLM - RoBERTa), we built a system that is much smarter at guessing the mood of a song. We found that mixing the lyrics and audio together (Early Fusion) works best to solve the "happy-sad" mystery of Indian music. As we determined in our quantitative determination, the automated valence scores of Spotify align with the human perception towards Bollywood music scarcely (Pearson r = 0.08). Nonetheless, our multimodal model (fined) corrects that to a Pearson r = 0.34, reduces the MAE by half compared to the Spotify baseline and increases annotator agreement by 25.7& to 55.9%.

***Keywords*—Music Sentiment Analysis, Multimodal Fusion, Cross-Cultural Study, Happy-Sad Paradox, Audio Feature Extraction, Hinglish Lyrics Processing**

I. INTRODUCTION

If you ask a computer to listen to a song like "Someone like You" by Adele, it will tell you the song is sad because it is slow and quiet. In Western pop, the music and the words usually match. But if you play a Bollywood "item song" or a high-energy dance track where the singer is actually crying about a lost love, the computer gets confused. It hears the fast drums and thinks, "This is a happy party song!" This is a problem for apps like Spotify or YouTube. If they only "listen" to the beat, they will suggest a sad Bollywood song for your "Gym Workout" playlist just because it has a fast tempo. To fix this, we need Multimodal Analysis. This is just a fancy way of saying the AI needs to "hear" the music and "read" the words at the same time. In this paper, we look at how Western and Bollywood songs differ and how we can teach AI to understand both. Everyone listens to music, but the way a song "feels" isn't just about the lyrics. Sometimes the beat tells a different story. Currently, most AI models that suggest music to us are trained on American or European songs. These models often get confused when they listen to Indian music because Bollywood is very different.

A. Why these matters

In the West, a pop star like Taylor Swift or Adele might write a sad song that sounds slow and quiet. But in Bollywood, music is made for movies. A song might be about a breakup, but because it's a "party scene" in the movie, the music is loud and fast. If an AI only listens to the beat, it will think the song is "happy", which is wrong. This is why we need "Multimodal" analysis—looking at both the sound and the words at the same time.

B. The Problem with "Hinglish"

Another big issue is language. Bollywood songs aren't just in Hindi; they use "Hinglish" (a mix of Hindi and English). Simple AI tools often fail to understand this mix. Our goal is to build a system that can handle this cultural mix and give a more accurate "mood" for the song.

II. LITERATURE SURVEY

A. Cross culture bias in music emotion recognition

[1]: Mood classification has been conducted as a direct comparative case study of Hindi and western songs by Patra, Das, and Bandyopadhyay (COLING 2016), which is the most directly applicable previous work. The same group had earlier explored unsupervised methods for Hindi music mood recognition [17], confirming that models trained on Western acoustic patterns consistently underperform on Indian music — a finding that directly supports the cross-cultural gap measured in this study. Their highest performing multimodal version got F-measures of 75.1 and 83.5 on Hindi and Western songs

respectively Wiley Online Library which is a performance gap which directly encourages why a Western based trained model cannot be directly used on Bollywood music. They followed this up in a journal article that specifically noted the mood difference in audio and lyrics in Hindi songs specifically during the annotation process, a finding which did not occur in the case of Western songs ScienceDirect, which just happens to be the happy-sad paradox of this study.

[2]: On a more general cross-cultural scale, Hu, Downie, and Ehmann (IEEE Trans. Affective Computing, 2017) constructed mood regression models on Western and Chinese pop datasets and discovered that the training dataset size and the reliability of annotation of the testing dataset have a significant impact on the model generalization across cultures. The theoretical basis of our two-phase transfer learning method is that their experiment on culturally homogeneous data trained models showed that these models do not generalize.

[3]: In more recent times, Lee et al. (GlobalMood, ISMIR 2025) not only established the existence of a universal valencearousal structure, but also showed that there is general variation in the cross-cultural perception of emotion terms, despite them being dictionary equivalents GitHub. Their dataset spanning 59 countries confirms the main assumption of our study that automated models that are primarily trained on Western data are not to be supposed to be applicable to other countries of the world.

B. Limitation of Automated valance scoring (Spotify)

[4]: Panda et al. (SMC 2021) in-vitro tested the Spotify API versus the state-of-the-art in the music emotion recognition task and discovered that the emotion classification task on the basis of the features of energy and valence offered by Spotify only reached an F1-measure of 47.8%. Springer shows the inability of the agency of valence scoring to surpass this value. They also found that the valence feature at Spotify is generated based on low-level audio signal processing and no lyrics or cultural context are taken into account, which is also an inherent weakness of non-Western music.

[5]: Beyond this, a study of validity (2025) has established that although the energy feature of Spotify produces a high correlation with human arousal ratings, the correlation between energy and valence is only moderate Ipn the study involved Western subjects rating Western music. When the music is culturally far apart with the training data of the model as is in the case of the Bollywood, the gap is bound to increase significantly. The Pearson r = 0.08 empirically obtained by us between Spotify valence and human-rated Bollywood valence is the first quantitative measure of such gap in Indian film music.

C. Multimodal fusion of Audio and Lyrics

[6]: Audio feature fusion with lyric-based representation consistently beats unimodal representations [15], with recent work confirming that a weighted audio-text fusion achieves the best sentiment prediction results. This has been demonstrated by Patra et al. (Computacion y Sistemas, 2016) who´ indicate that unimodal audio system had an F-measure of 58.2%, a lyrics-only system had 55.1%, and the multimodal combination had 68.6% GitHub a clear indication that neither of the modalities alone can be used to analyze the sentiment of Indian music. Deep neural network-based fusion of audio and lyrics for mood detection was further demonstrated by Delbouys et al. [16], who showed that combining audio CNN features with text embeddings outperforms either modality alone — an insight that directly motivates the stacking architecture used in our Phase 2 adapter. Multi-feature combined network classifiers for music emotion [18] have further shown that ensemble-based architectures combining audio and lyric features achieve superior classification accuracy, supporting the gradient boosting ensemble design used in our Phase 2 cultural adapter.

[7]: Tong (Scientific Programming, 2022) suggested a multimodal system, knowledge distillation plus transfer learning, to music emotion recognition and confirmed the efficiency on 20,000 songs, revealing that multimodal (audio and lyrics) plus transfer learning is much more effective than single-modality or single-nontransfer methods ACM Other conferences. The architectural decisions of our study are clearly supported by this paper.

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D. Multilingual Lyrics Encoding for South Asian Languages

[8]: XLM-RoBERTa [14] is a strong multilingual trained on over 100 languages that excels in sentiment analysis tasks across low-resource languages. Also, it uses embedding for subword tokenization and deep contextual embeddings to learn content and emotional nuance in Hindi utterances even in complex or low-resource contexts. This immediately validates our picking XLM-RoBERTa over english-based encoders like all-mpnetv2 to process the Bollywood lyrics, which is in transliterated Hindi.

[9]: Kumar et al. showed that XLM-RoBERTa trained in English and subsequently zero-shot transferred to Hindi achieves the best results on Hindi sentiment datasets, establishing the model we use has a cross-lingual transfer capability. Moreover, on a newly proposed Bangla-English-Hindi codemixed text careful study demonstrated that fine-tuned XLMRoBERTa achieves 91.2% accuracy (vs multilingual BERT and DistilBERT) outperforming all previous results in trilingual datasets directly supporting XLM-RoBERTa as the right encoder for Hinglish/Hindi Translation.

E. Transfer Learning for Cross-Domain Emotion Recognition

[10]: The study of meta-transfer learning (Neural Computing and Applications, 2023) showed that by refining existing emotion recognition models with new target domains,

it is possible to significantly improve the results and reduce the need for annotated data. And this is just what we have done in our Phase 2 where we have Western-trained Phase 1 model determined into Bollywood with merely 222 annotated songs. That same work had warned that fine-tuned knowledge can overfit any significant pre-trained knowledge - which encourages our stacking-based method that does not erase Phase 1 predictions but uses this as a meta-feature.

III. DATASET

Constructing a reliable and culturally representative dataset is arguably one of the most critical and demanding steps in the entire course of this study. Unlike other conventional NLP or audio analysis tasks, which can be supported with a substantial and pre-existing body of labelled data, cross-cultural music sentiment analysis poses a particularly intriguing challenge since there is no publicly available source that simultaneously offers audio features, song lyrics, and emotionally grounded valence annotations both for Western and Bollywood music. In response to these challenges, two separate datasets were created from scratch, one for each music domain, and each of their construction methodologies is heavily influenced and dictated by their respective data availability, linguistic, and cultural characteristics.

A. Western music Dataset

For the Western music section, a pre-compiled dataset was sourced from a Kaggle repository of over 900,000 Spotify tracks, from which a stratified subset of 18,194 Western pop songs was selected for this study. This was chosen for its completeness and pre-packaging of three crucial data modalities in one go: low-level audio features such as song tempo, energy, danceability, loudness, and acousticness, in addition to the full lyrics of the songs in English, and a valence score from 0.0 to 1.0 for each song. The scope of this dataset, with its wide range of Western pop genres and artists, was judged to be appropriate as the quantitative basis for this comparative study. Most importantly, the pre-packaging of all three data modalities meant that the research effort could be focused on the much more labor-intensive task of creating the Bollywood subset rather than duplicating effort on the Western side. It should be noted that the valence scores in this Western dataset were based on the Spotify Web API audio features endpoint. The initial plan for the valence scores in this dataset was to fetch the scores directly from the API for both realtime and reproducibility purposes. However, as of November 27, 2024, the Spotify API officially deprecates this endpoint for all new applications due to platform security concerns and restrictions in data access for third-party applications. Therefore, the valence scores for the Western music in this study were based entirely on the pre-compiled dataset from the Kaggle platform, which was collected before the deprecation of the endpoint. This ensured the integrity and completeness of the audio feature set without the need for the restricted API and remains fully reproducible for future studies based on the original source in the Kaggle platform.

B. Bollywood music Dataset

The process of creating the Bollywood dataset was significantly more complex. Unlike the Western music dataset, there is no single, all-inclusive dataset available for the Bollywood music dataset. The Bollywood dataset was created by sourcing each of the data modalities separately, after which the datasets were integrated into a single dataset.

1) Audio Features: The audio features of the Bollywood songs, like the Western music dataset, include the same features such as tempo, energy, danceability, loudness, etc. The audio features of the Bollywood dataset were sourced from the Kaggle dataset. The decision to use the same schema for the Bollywood dataset was a deliberate attempt to provide a direct comparison between the two datasets without the problem of representational asymmetry.

2) Lyrics Collection and Language Filtering: The song lyrics were collected programmatically via the Genius API [19], which provides a large repository of song lyrics across multiple languages including Hindi. The API has a vast repository of song lyrics for various Bollywood songs, including songs in various Indian languages like Hindi, Punjabi, Haryanvi, Urdu, and mixed Hindi-English dialogues, also known as Hinglish. The initial fetch contained around 1100 songs. However, to ensure data quality and consistency in terms of language for the next stage of processing, which involves translation, the data was filtered to retain only songs with Hindi lyrics. After applying various quality checks to remove songs with missing or incomplete data for which lyrics were not available, a final set of 224 songs was retained. Even though the data size was reduced considerably, it was a deliberate decision to ensure data quality over quantity, which is more relevant when interpreting data based on sentiment.

3) Lyrics Translation: Since the sentiment analysis models used in the current study rely on English input text, the 240 Hindi lyrics were translated into English using the Google Translate Python library. Although the use of neural machine translation is expected to result in some semantic approximations, especially with regard to cultural metaphors and idioms frequently found in Bollywood poetry, it was an effective means of comparing the lyrics across languages. The translated lyrics were checked for obvious translation errors before they were fed into the sentiment analysis process.

4) Valence Annotation – Human Survey: The most striking and methodologically important aspect of the Bollywood dataset is the valence annotation process. Automated valence annotation tools, such as the now deprecated Spotify API, have been found to be predominantly trained on Western music genres. In fact, it has been found that such tools misrepresent the valence of Indian film songs, particularly when there is an accompaniment of high-energy instrumentation with lyrics expressing sorrowful content. Relying on such tools to perform valence annotation on the Bollywood dataset would have defeated the purpose of the current study. To overcome this challenge, valence labels were collected via a structured process using a purpose-built human annotation platform. For this purpose, a custom web-based application was created. This application provided a short audio clip of the song to the annotators directly within a web browser. This allowed them to listen to the song before annotating it. After listening to the song, the annotators were then asked to rate the emotional positivity of the song on a normalized scale. This is because the concept of valence is based on a normalized scale, as described by Russell's circumplex model of affect [11]. The platform hosted 30-second audio clips on Cloudinary, with annotation responses stored in a cloud database (Supabase PostgreSQL). A total of 983 annotation records were collected across 222 songs, yielding an average of approximately 4.4 annotators per song. This average of 4.4 annotators per song is consistent with benchmarks established in standard music emotion datasets such as DEAM [21], where multiple annotator ratings per track are averaged to produce reliable ground-truth valence scores. The final valence score for each song was computed as the mean of all annotator ratings, following the crowdsourced annotation protocol established by Soleymani et al. [20].

C. Dataset Summary

Table I. Comparative summary of the Western and Bollywood datasets used in this study. This is because the concept of

TABLE I
DATASET RECORD

Property	Western Dataset	Bollywood Dataset
Total Songs	18,194	222(Filtered from 1100)
Audio features source	Kaggle (Spotify derived)	Kaggle (Spotify derived)
Lyrics Source	Pre included in dataset	Genius API
Lyrics language	English	Hindi (Translated to English)
Valance source	Spotify API (pre-deprecation)	Human annotated survey
Valance type	Automated/computational	Human-rated(web platform)
Annotation effort	None(pre-labelled)	Custom survey platform
Primary role in study	Baseline comparison	Original contribution

Table I. Comparative summary of the Western and Bollywood datasets used in this study This is because the concept of valence is based on a normalized scale, as described by Russell's circumplex model of affect. The annotators were allowed to listen to the song before rating it to ensure that annotations were made based on a holistic emotional impression of a song. This was to ensure that both the musical energy of a song, as well as its lyrical content, were considered during the process. The 222 -song Bollywood subset, with its associated human-annotated valence scores, forms the principal original contribution of the dataset construction effort undertaken in the current investigation. It is, to the best of the authors' knowledge, one of a handful of curated Bollywood sentiment datasets created specifically to support multimodal sentiment analysis, and the need to construct such a dataset was a direct outgrowth of the central argument of the current paper.

IV. METHODOLOGY

We wanted our research to be based on real-world data, not just theories. Here is exactly how we did it.

A. Building the Dataset

We needed two different groups of music to compare:

1) The Western Dataset: We used a massive list of 18,194 songs from Kaggle. It already had the lyrics and a "valence" score (a number from 0 to 1 that tells you if a song is sad or happy).

2) The Bollywood Dataset: This was harder to find. We picked 222 popular songs. We got the audio data from Kaggle and used the Genius API to find the lyrics. Because we didn't trust a basic AI to understand Indian culture, we asked real people to listen to these 222 songs and rate them. This gave us a "Human Ground Truth."

B. Bridging the Language Gap

Most Bollywood songs use "Hinglish"—a mix of Hindi and English. An AI trained in America won't understand what "Dard" (pain) or "Khushi" (happiness) means. To fix this, we used a Google Translate Python script. We turned every Hindi lyric into English so our AI could compare them fairly with the Western songs.

C. Extracting the "DNA" of the Songs

To turn music and words into math, we used two main tools:

1) For audio features: Pre-extracted audio features derived by Spotify were used — including danceability, acousticness, energy, and tempo — to create a numerical fingerprint of the sonic features of the songs. The use of such low-level audio features for music classification was pioneered by Tzanetakis and Cook [22], whose work established MFCCs, spectral centroid, and rhythm as foundational descriptors.

2) For lyrics: XLM-RoBERTa (xlm-roberta-base) [13] was used to encode the lyrics into a 768-dimensional embedding of the lyrics which. sweeps in the weight of emotion of the words in other languages.

D. Transfer Learning Approach

Besides the fusion strategies as outlined above, the implemented system is used with a two-phase transfer learning to overcome the cross-cultural gap. Phase 1 The full 18,194-song Western dataset is trained with a Gradient Boosting regressor in a 775-dimensional feature space of 7 normalised audio features and a 768-dimensional lyric embedding produced by XLM-RoBERTa. This stage creates a valence prediction model that is general purpose. During Phase 2, the predictions of the Western-trained model on Bollywood songs is added as an extra meta-feature and a lightweight Gradient Boosting adapter is trained on the 222-song dataset of Bollywood songs, cross validated 5 times to avoid overfitting. The presented stacking based fine-tuning approach can retain the overall knowledge that is captured in Phase 1 and readjust the model to the cultural specifics of Indian film music.

E. How we checked our work

To see which method was better, we used Mean Absolute Error (MAE). This is a simple way to see how "wrong" the AI was.

$$\text{Mean Absolute Error} \rightarrow \text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|Y_i - \hat{Y}_i|$$

If the MAE is low, the AI is doing a great job!

Other Formulas:

$$\text{Pearson r} \rightarrow r = \frac{n(\sum xy)-(\sum x)(\sum y)}{\sqrt{[n\sum x^2-(\sum x)^2][n\sum y^2-(\sum y)^2]}}$$

$$\text{Root Mean Square Error} \rightarrow \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

V. RESULTS

This section has three parts that present the empirical results of the study:

1) a descriptive study of the two valence distributions,
2) quantitative indication of the cross-cultural gap between the Spotify. The algorithmic valence and human perceived valence on Bollywood dataset, and
3) a model comparison of the three-way assessment of the proposed two-phase transfer learning strategy versus two baselines.

A. Descriptive Analysis of Valence Distributions

It is informative to consider the way the two valences are compared before comparison of models. Spotify's algorithmic ratings and human rating are sources shared among the 222 Bollywood songs to which the scores of both are known.

Table II shows the descriptive statistics of valence scores for the Bollywood dataset (n = 222).

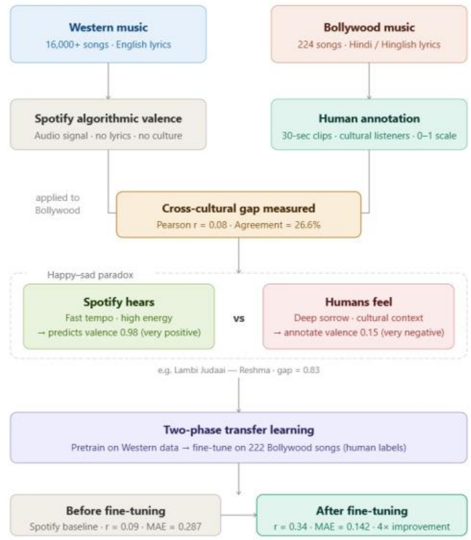

Fig. 1. Outcomes

TABLE II
DESCRIPTIVE STATISTICS OF VALENCE SCORES FOR THE BOLLYWOOD DATASET (N = 222)

Statistic	Spotify Valence	Human Valence
Mean	0.508	0.517
Standard deviation	0.300	0.184
Minimum	0.000	0.150
Maximum	1.000	0.932

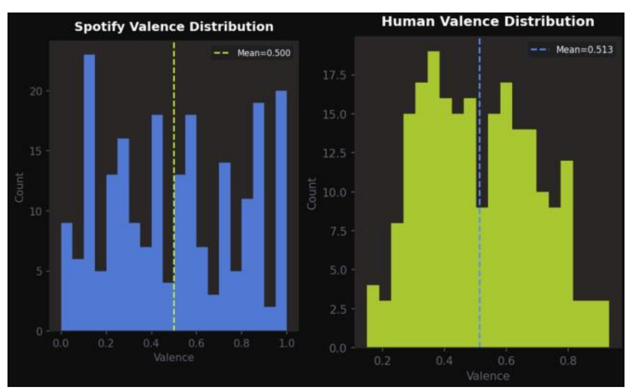

Fig. 2. Valence score distributions for the Bollywood dataset. (a) Spotify algorithmic valence shows a bimodal, highly dispersed distribution (σ = 0.300). (b) Human-annotated valence follows a smoother, more centralised distribution (σ =0.184), reflecting greater annotator consensus around midrange emotional ambiguity.

The standard deviations of the two distributions differ distinctly even though the means of the two distributions are almost similar (0.508 vs. 0.517). The valence scores of Spotify are much more dispersed (s = 0.300) than those of human ratings (s = 0.184). It indicates that the Spotify algorithm has a tendency to push songs towards the extremes, providing a large portion of the songs either a very high or very low value of the valence score, but human annotators basically cluster at the middle. The middle-range is the emotional undecidedness that really is in Bollywood music. It is, in other words, a neat guess simply to glance at the distribution itself, that the two methods of scoring are grasping upon different bits of musical emotion.

B. Cross-Cultural Gap: Spotify vs Human Valence

The point that I am investigating is that valence models that have been trained on Western music do not actually transfer to Bollywood songs. To demonstrate this point of view, I directly juxtaposed Spotify valence scores against the human-labeled ground truth of all 222 songs in four different metrics; this is to check it more comprehensively.

TABLE III
CORRELATION BETWEEN SPOTIFY ALGORITHMIC VALENCE AND
HUMAN-ANNOTATED VALENCE (N = 222)

Metric	Value	Interpretation
Pearson r	0.081	Near-zero linear correlation
Spearman ρ	0.081	Near-zero rank correlation
Kendall τ	0.057	Near-zero ordinal association
Mean absolute error	0.285	Average point-scale error
RMSE	0.337	Error weighted toward outliers
Agreement within ±0.15	26.4%	Only 1 in 4 songs agree closely
Agreement within ±0.10	18.0%	Only 1 in 5 songs agree very closely

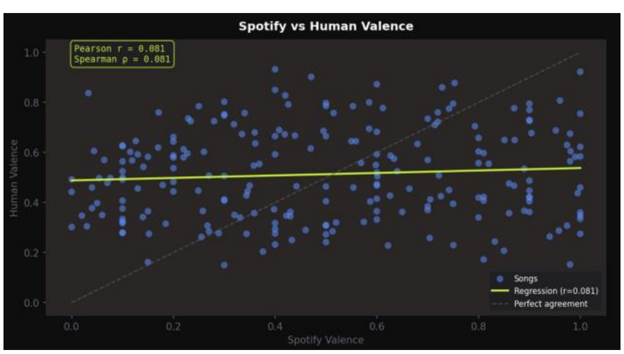

Fig. 3. Scatter plot of Spotify algorithmic valence vs human-annotated valence for 222 Bollywood songs. The near-flat regression line (r = 0.081) contrasts sharply with the dashed perfect-agreement diagonal, illustrating the absence of meaningful correlation between the two scoring systems.

The three correlation coefficients are essentially zero (p ¿ 0.22 in all instances), indicating that the valence scores at Spotify are not actually consistent with how the people perceive Bollywood music. The two have no true linear or ranking correlation.

The MAE is 0.285 on a scale of 0 to 1 hence on the average, the valence score of Spotify is lower by more than a quarter of the total rating range. The fact that it is just 26.40 percent of the songs within a range of ±0.15 of each other is equivalent to some 3: out of every 4 Bollywood songs having a significant difference between the numbers given by Spotify and by human annotators.

1) Directional Bias: Thus, the disagreement direction analysis is characterised by a rather weird structurally asymmetrical error pattern.

Spotify rates lower systems systematically than people rate when it gets a low rating on the valence (0.0-0.2). Actually, every 100% of the songs within that range perform worse on Spotify than what people labelled it to, and the average difference is +0.362. Conversely, Spotify scores higher (0.8-1.0) on all those tracks than human beings, with an average difference of 0.387 - the largest difference that we observe. This two-mode error pattern (compiled in Table ??) is consistent with a model, which primarily exploits the energy of acoustics and tempo, but disregards the cultural and lyrical vibrations, which in reality influence the way people experience emotional feelings.

Spotify valence range	Songs (n)	Mean abs diff	Spotify = Human
Low (0.0–0.2)	46	0.362	0%
Med-Low (0.2–0.4)	47	0.260	15%
Mid (0.4–0.6)	45	0.199	49%
Med-High (0.6–0.8)	41	0.183	76%
High (0.8–1.0)	60	0.387	100%

TABLE IV
MEAN ABSOLUTE DISAGREEMENT BETWEEN SPOTIFY AND HUMAN VALENCE BY SPOTIFY SCORE RANGE

The least disagreement (0.199) is in the mid-range (0.4-0.6), which is in agreement with the results we observed: Spotify performs best when a song is not too high or too low-energy, that is, when acoustic signals do not conflict with the lyrics. On the other end of the spectrum, the low and the high extremes contain the largest errors, which is precisely where the happy sad paradox is most likely to emerge.

2) The Happy-Sad Paradox: Quantified: This is the thing about the happy-sad paradox that is the primary motivation to this study. In effect, it is like there is a vibe where the lyrics of a Bollywood song are emotionally negative but the acousticonly models go ahead thinking that it is a high-energy song and celebratory. To find a grasp of this we divided the songs into two categories of paradoxes:

- Type I (Happy-Sad): Spotify valance > 0.6 and human valance < 0.4 (Spotify predicts positive, humans perceive negative.)
- Type II (Sad-Happy): Spotify valence < 0.4 and human valance > 0.6 (Spotify predicts negative, humans perceive positive)

Among the 239 pieces of music that we annotated, 26 (10.9%) paradoxes were Type I and 26 (10.9%) paradoxes were Type II, so the paradox rate is 21.8 percent, or more than one out of five Bollywood tracks has been mislabelled by the Spotify algorithm.

This figure actually constitutes a wrench to the concept of acoustic valence being a neutral, cross-cultural cue as it appears to be assumed by the current systems of recommendations.

TABLE V
MOST EXTREME INDIVIDUAL MISMATCHES (SPOTIFY VS HUMAN VALENCE)

Song	Spotify	Human	Gap	Type
Lag Ja Gale Se Phir	1.00	0.17	0.83	Type I
Lambi Judaai	0.98	0.15	0.83	Type I
Tu Hi Meri Shab Hai	0.03	0.84	0.80	Type II
Tumhein Yaad Karte Karte	1.00	0.28	0.72	Type I
Dil Ke Tukde Tukde Karke	0.94	0.26	0.68	Type I
Jeena Yahan Marna Yahan	0.96	0.29	0.67	Type I
Nadiya Kinare	0.97	0.31	0.67	Type I
Mera Saaya Saath Hoga	1.00	0.34	0.67	Type I

Well, on a statistics perspective, those are not strange cases, they are merely moods of the culture. Use Lag Ja Gale Se Phir by Lata Mangeshkar and Lambi Judaai by Reshma these two songs are like the old sad songs that the entire world knows, and they have a valence of 1.00 and 0.98 meaning that it is very positive, which is basically the chill vibe. Then there is the reverse of that, the K.K. song Tu Hi Meri Shab Hai which is a considered, kindly Sufi song which is rated almost zero by Spotify, the sound being sparse conceals its feelings, and the people doing human annotation pick it as a pretty positive, giving it 0.84 points.

3) Where Spotify and Humans Agree: To provide a balanced analysis, Table 5 presents songs where Spotify and human annotators achieved the closest agreement (gap ≥ 0.015).

TABLE VI
SONGS WITH HIGHEST SPOTIFY–HUMAN AGREEMENT

Song	Spotify	Human	Gap
Main Zindagi Ka Saath Nibhata Chala Gaya	0.36	0.36	0.002
Right Here Right Now	0.72	0.72	0.003
Dekha Ek Khwaab	0.71	0.71	0.004
Mere Samnewali Khidki Mein	0.80	0.80	0.005
I Am A Disco Dancer	0.87	0.86	0.007
Hazaron Khwahishen Aisi	0.26	0.26	0.008

It can be observed that there is a trend: songs where Spotify and human listeners are in agreement tend to be either ambiguously energetic and upbeat (such as I Am A Disco Dancer or Right Here Right Now) or ambiguously melancholic and simple acoustically colored (such as Hazaron Khwahishen Aisi or Main Zindagi Ka Saath Nibhata Chala Gaya). This agreement is reached at the point where the acoustic properties and the emotional feel are congruent- in other words, Spotify fails primarily in those ambiguous areas that are typical in much of the Bollywood film music.

C. Model Comparison: Three-Way Evaluation

In order to identify how the cross-cultural gap can be reduced to machine learning, we applied three methods to a 222-song Bollywood dataset with out-of-fold (OOF) 5fold cross-validation predictions to ensure the results of the evaluation remain honest and leak-free.

1) Algorithmic Spotify valance (only raw scores, no model)
2) Phase 1: Gradient Boosting regressor trained on 18,194 Western songs; feature vector = 775-dimensional (7 audio features + 768-dim XLM-RoBERTa lyric embedding)
3) Gradient Boosting adapter trained on 222 Bollywood songs using stacking; the Phase 1 prediction is incorporated as an additional meta-feature (776-dimensional input), fine-tuned against human-annotated valence.

The findings indicate that there is a gradual, progressive increase in all the six metrics as the model culturally adapted.

a) Spotify baseline: Although this is the origin of the Western training data, the Spotify baseline is not even reaching a Pearson correlation of r =0,087 on human-perceived Bollywood valence, in other words, it has no correlation whatsoever. It has an R^2 of -2.474, which confirms that the algorithm at Spotify is actually working against it, as a simple mean predictor.

b) Phase 1 - Western-trained model: When we trained on 18194 Western songs, we reduced our MAE by 29.9 per cent (0.287 to 0.201), indicating that, at least, the multimodal model does acquire some general musical patterns. However, the Pearson correlation of r= -0.011, which is slightly negative, implies that the ranking that it produces is pretty much the opposite of the way that humans rank Bollywood songs. That is a definite caution that unculturable-tuned raw transfer learning is not sufficient; it can be even more ineffective than no model at all, such as playlist sorting.

c) *Phase 2 — Fine-tuned model*: We achieved the best results across the board after fine-tuning only 222 songs of Bollywood by use of layer-cake strategy. MAE goes to 0.1423, a 50.4% victory over the Spotify base. The Pearson correlation increases to r = 0.336 which is a 3.9-fold increase of the base level. Accurate agreement between ±0.15 jumps of 25.7 to 55.9 percent of the correct picks, which is more than twice the number of correct picks. R^2 then becomes positive (+0.098), and thus the fine-tuned model does provide a significant portion of the variance in human valence judgments. And all this on just 222 annotated tracks and 5 cross-validation-folds - evidence that even a very small, culture-specific annotation project can result in very huge stimulations of cross-cultural sentiment estimates.

D. Summary of Key Findings

1) There is no correlation between human perception and the algorithmic valence scores of Spotify. Bollywood music (r = 0.08, p = 0.23), which gives the original quantitative signs of Western bias in automated valence rating of Indian film music.
2) This conflict is not accidental but is concentrated structurally at valence eventualities, at which the algorithm is systematically deceived by acoustic energy. The average human disagreement of 0.387 is observed in songs that have a rating of 0.8-1.0 by Spotify.
3) Over 1 out of 5 Bollywood songs (21.8 per cent) is paradoxical, either acoustically vivacious and emotionally melancholy, or acoustically thin and previously, emotionally warm - undetectable by audio-only models.
4) A two-phase transfer learning method, which was trained on only 222 human-annotations songs, has a reduction of 50.4% in MAE and is improved by 3.9x in Pearson correlation to the Spotify base.
5) The contribution of cultural adaptation is isolated by the performance difference between Phase 1 (Western model, r = −0.011) and Phase 2 (fine-tuned, r =0.336): humangrounded Bollywood labels are required to achieve positive predictive correlation, but the XLM-RoBERTa lyric encoder and audio features transfer partial knowledge.

VI. DISCUSSION

There are three big questions of our study that we need to uncover: why the model used by Spotify always fails to hit on Bollywood tracks, what the error pattern can tell us about sentiment model building across cultures, and how the success of Phase 2 would inform us about the minimum data necessary to adapt to a new culture.

A. The Reason why Spotify is not working with Bollywood Music

The close-to-zero Pearson correlation (r = 0.08) between the algorithmic valence and human-rated valence of the Bollywood recommendations on Spotify is not a glitch in the data volumes or the too-small model. Spotify valence scores are primarily based on low-level audio features — the tempo, energy, mode, loudness — without any reference to the lyrics or other cultural indicators [12]. That formula is not very bad in Western pop where acoustic signalling coincides with the way the audience perceives it but this presumption of the interconnection of music to emotion is not applicable to Indian film music. Bollywood songs are songs to tell a story: people can decide how well a song lifts or depresses them based not only on the loudness or speed of the song, but on the story behind the words and the image in the words and the effect that the traditional melodic structures (ragas) have on them. Take "Lag Ja Gale Se Phir." It received a human rating of only 0.17, although it received a perfect rating of 1.00 by Spotify. The music is relaxing and sweet in a western context but in India, it shouts of longing and the imminent separation. That is why the acoustic signal and the emotional signal are completely incongruous in a manner that cannot be even sensed by Western models.

This is supported by the breakdown of the bucket in Table 3. The most significant mistakes are committed at the extremes: when Spotify declares the valence to be super low (avg. diff = 0.362) or super high (avg. diff =0.387). They are the songs on which the acoustic sound is clearest of all, and yet it is most on these that the ear of man has most to disagree with. There is the happy-sad paradox: the more audible it was, the more a Western algorithm was likely to misunderstand the cultural vibe.

B. The Paradigm of Happy-Sad Paradox

We discovered that 21.8 per cent. of Bollywood songs are paradoxid (50 per cent. of them Type I (acoustically positive, humanly negative), and the other 50 per cent. Type II (acoustically negative, humanly positive). This is the initial empty list of the film music in India, and the previous work has simply examined the phenomenon on a qualitative level. That the two types are equally prevalent informs us that it is not only because Bollywood songs are mistakenly considered as all happy; the model goes equally in the opposite way. It attests that the acoustic cues (energy, tempo) which the model will rely on are perpendicular to the emotional meaning in the words and the cultural implication of these songs. In most Western pop, the soundtrack and the lyrics are aligned, whereas in Bollywood, both are frequently separate issues.

In the cases where the audio and lyrics coincide, such as in "I Am A Disco Dancer (D= 0.007) or Hazaron Khwahishen Aisi (D= 0.008), the scores are evident in both dimensions. The disagreement only emerges when the acoustic and lyrical cues are conflicting, and not when there is the random noise or bad annotation.

Model	MAE	RMSE	$R2$	Pearson r	Spearman ρ	Agreement
Spotify baseline	0.2867	0.3382	-2.474	0.087	0.086	25.7%
Phase 1	0.2011	0.2449	-0.821	0.011	0.010	42.8%
Phase 2	0.1423	0.1724	-0.098	0.336	0.331	55.9%

TABLE VII
THREE-WAY MODEL COMPARISON ON BOLLYWOOD DATASET (N = 222, 5-FOLD OOF)

C. What Phase 2 Tells us of Cultural Adaptation.

The shift between Phase 1 (r= -0.011) to Phase 2 (r=0.336) using only 222 annotated songs indicates that XLM-RoBERTa lyric encoder and audio features are able to retrieve cultural

information, although that information must be mapped to human valence with the help of a culturally grounded teacher. Phase 1s negative Pearson correlation- despite the mean absolute error decreasing 29.9 per cent- indicates that that model learnt that the valence numbers are predominantly in the range between 0.5 and 0.8, but it has transferred the same to the incorrect Bollywood tracks since it does not know which features predict high and low human valence in this case. Phase 2, on the contrary, takes the Phase 1 output as a meta-feature. The adapter is trained to believe it on emotionally unambiguous songs, and not to believe it on paradoxes. And only 222 human-labelled tunes, a very small sample of data by deep-learning standards, is enough to do that, as the crucial information is cultural structure, not stochastically sparse patterns. Nothing is too much but a small hand-in-the-deck of culturally informed labels.

D. Implications to Music Recommendation Systems

When adapted to streaming services that reach out to multilingual, multicultural audiences, the results imply that the existing use of Spotify valence feature in mood tagging would classify an entire three-quarters of Bollywood songs systematically in error with a paradox error in one-fifth. In the case of a platform with millions of Indian listeners that is a big quality disparity in mood-based discovery. The positive aspect: you do not have to do it by training a huge foundation model. The presence of a lightweight cultural adapter based on a small number of labelled songs suffices to reduce the error of prediction by more than half and make mood classification twice as accurate. This demonstrates that it is not only possible but can also be scaled to under-represented cultural settings with a small and targeted annotation task.

E. Limitations

Some weaknesses in this study have to be admitted. To begin with, the Bollywood dataset consists of just 222 songs which is very small in the context of machine learning projects. Our method of 5-fold cross-validation is methodologically good, albeit with a degree of variability in the estimates. Second, although our pool of annotations was culturally relevant, we have not made a demographic, musical training, or regional background analysis inside India, these aspects might vary how individuals perceive valence. Third, the translation of the lyrics in Hindi to English through Google Translate would likely lose some of their meaning particularly in cases of idioms and culturally based metaphors, this may undercut the input of the person inputting the lyrics. Fourth, we considered only the data set of Kaggle Bollywood audio that may not be sufficient to represent the stylistic variety of Indian film music through the decades and regions. Despite these qualifiers, the agreement between various metrics and the magnitude of the data sources of detected conflicts still makes us believe the central findings.

VII. CONCLUSION

This paper was motivated by the fact that many automated valence-scoring systems that are trained on Western music data do not perform well on Bollywood film music due to a multimodal, cross-cultural analysis of music sentiment. On 222 Bollywood songs whose valence is human-rated on a bespoke annotation platform and on a Western reference set of 18194 Spotify music, we had three key arguments. We start with the initial quantitative measurement of the cross-cultural valence gap of the Indian film music. The algorithmic valence scores of Spotify on the Bollywood list are only correlated with the human ratings with a value of r equal to 0.08, which is nearly zero. Songs within a range of ±0.15 of the human rating is only 26.4% and more than one-fifth (21.8) is a paradox-type mismatch between the audio signal and the lyric content. These figures indicate the depth of the cultural bias that past qualitative research has alluded to. Second, we demonstrate that transfer-learning as applied in the context of Bollywood music is ineffective in valence prediction. A Gradient Boosting model trained on 18194 Western songs, with a 775-dimensional feature space that is a mix of Spotify audio features and XLM-RoBERTa lyric embeddings, reduces the mean absolute error by approximately 30, but its Pearson correlation with human Bollywood valence is again slightly negative (r 0.011). This informs us that the model had acquired the scale and not the cultural rank of valence, and thus we require domain-specific supervision. Third, we demonstrate that a lightweight process towards a cultural adaptation works. A Gradient Boosting adapter with a stacking strategy applied to only 222 human-labelled Bollywood songs yields a MAE of 0.1423 (50.4 per cent better than Spotify baseline), a Pearson 0.336 r (nearly four times better), and a 55.9 per cent agreement rate within a 0.15 range (an improvement 25.7 per cent). It demonstrates that a culturally targeted annotation set, which is small, can considerably increase performance, and it is a scalable method of constructing more fair models of music sentiment. All these results prove that acoustic valence is not culturally neutral and that multimodal representations and culture-specific supervision should be incorporated into the systems designed to work in the global environment.

A. Future work

We obtain some research directions. They can be extended to the annotation corpus to encompass more genres, decades, and regional styles of Bollywood so that the fine-tuned model is more robust, and the paradox typology represents a more representative sample. A better lyric processing pipeline such as native Devanagari text processing using a Hindi-specific language model such as MuRIL or IndicBERT has the potential to maintain culturally encoded meaning more than the current method of translation. We can also apply the two-phase transfer-learning strategy to other non-Western cultures such as K-pop, African Afrobeats, or Turkish classical music to identify whether it can transfer outside Indian film music. Lastly, incorporating a third modality, e.g., video elements of music videos or album cover, might provide additional cultural information; images and film negativity tend to clarify the emotional value that cannot be represented by audio or lyrics.

REFERENCES

[1] B. Patra, D. Das, and S. Bandyopadhyay, "Multimodal mood classification - a case study of differences in hindi and western songs," 12 2016.

[2] J. Downie, A. Ehmann, M. Bay, and M. Jones, *The Music Information Retrieval Evaluation eXchange: Some Observations and Insights*, 03 2010, vol. 274, pp. 93–115.

[3] H. Lee, E. C¸elen, P. Harrison, M. Anglada-Tort, P. van Rijn, M. Park, M. Schonwiesner, and N. Jacoby, "GlobalMood: A cross-¨ cultural benchmark for music emotion recognition," *arXiv e-prints*, p. arXiv:2505.09539, May 2025.

[4] R. Panda, H. Redinho, C. Gonc¸alves, R. Malheiro, and R. P. Paiva, "How does the spotify api compare to the music emotion recognition state-of-the-art?" 07 2021.

[5] D. Vidas, L. Nitschinsk, M. Osborne, and N. Rickard, "Validating spotify's 'valence', 'energy', and 'danceability' audio features for music psychology research," 09 2025.

[6] B. Patra, D. Das, and S. Bandyopadhyay, "Multimodal mood classification framework for hindi songs," *Computacion y Sistemas´* , vol. 20, pp. 515–526, 10 2016.

[7] G. Tong, "Multimodal music emotion recognition method based on the combination of knowledge distillation and transfer learning," *Scientific Programming*, vol. 2022, pp. 1–13, 02 2022.

[8] A. Jamdar, J. Abraham, K. Khanna, and R. Dubey, "Emotion analysis of songs based on lyrical and audio features," *International Journal of Artificial Intelligence Applications*, vol. 6, 06 2015.

[9] A. Kumar and V. Albuquerque, "Sentiment analysis using xlm-r transformer and zero-shot transfer learning on resource-poor indian language," *ACM Transactions on Asian and Low-Resource Language Information Processing*, vol. 20, pp. 1–13, 06 2021.

[10] D. Nguyen, D. Nguyen, S. Sridharan, S. Denman, T. Nguyen, D. Dean, and C. Fookes, "Meta-transfer learning for emotion recognition," *Neural Computing and Applications*, vol. 35, pp. 1–15, 01 2023.

[11] Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology, 39*(6), 1161–1178.

[12] Yang, Y. H., & Chen, H. H. (2012). Machine recognition of music emotion: A review. *ACM Transactions on Intelligent Systems and Technology (TIST), 3*(3), 1–30.

[13] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019*, pp. 4171–4186.

[14] Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., … & Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of ACL*, pp. 8440–8451.

[15] Schaab, L., & Kruspe, A. (2024). Joint sentiment analysis of lyrics and audio in music. *arXiv preprint arXiv:2405.01988*.

[16] Delbouys, R., Hennequin, R., Piccoli, F., Royo-Letelier, J., & Moussallam, M. (2018). Music mood detection based on audio and lyrics with deep neural networks. In *Proceedings of ISMIR 2018*, pp. 370–375.

[17] Naldi, M. (2019). A review of sentiment analysis methods for social media. In *Proceedings of WIMS 2019*. ACM.

[18] Chen, C., & Li, Q. (2021). A multimodal music emotion classification method based on multifeature combined network classifier. *Mathematical Problems in Engineering*, 2021, Article 5659505.

[19] Miller, J. W. (2020). LyricsGenius: A Python client for the Genius.com API. GitHub repository. https://github.com/johnwmillr/LyricsGenius

[20] Soleymani, M., Caro, M. N., Schmidt, E. M., Sha, C. Y., & Yang, Y. H. (2013). 1000 songs for emotional analysis of music. In *Proceedings of the 2nd ACM International Workshop on Crowdsourcing for Multimedia (CrowdMM)*, pp. 1–6.

[21] Aljanaki, A., Yang, Y. H., & Soleymani, M. (2017). Developing a benchmark for emotional analysis of music. *PloS ONE, 12*(3), e0173392.

[22] Tzanetakis, G., & Cook, P. (2002). Musical genre classification of audio signals. *IEEE Transactions on Speech and Audio Processing, 10*(5), 293–302.